

Total no. of Pages 02

Roll no

3rd Sem.-B.Tech.

END-SEMESTER EXAMINATION, NOV.-DEC.,2024

Course Code- EAMTC04

Course Title- Mathematics for Machine Learning

Time- 03 Hours

Max. Marks-50

Note: - Attempt all five questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer.

Q.No.	Attempt any 2 parts of the questions below	Marks	CO
1(a)	Find KerT & ImT for the linear mapping $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (0, y)$. Verify that $\dim(\text{Ker}T) + \dim(\text{Im}T) = \dim R^3$	05	CO1
1(b)	Find rank and nullity of the linear transformation $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x + y, 2z - x)$.	05	CO1
1(c)	Examine if the following set S is a subspace of R^3 (i) $S = \{(x, y, z) \in R^3 : x - 10 = 0\}$ (ii) $S = \{(x, y, z) \in R^3 : x + 2y + 3z = 0\}$	05	CO1
2(a)	Use the Gram-Schmidt process to convert the basis $\{(1, 2, 2), (-1, 0, 2), (0, 0, 1)\}$ of the Euclidean space R^3 with standard inner product into an orthonormal basis.	05	CO2
2(b)	Find the eigen values and eigen vectors of $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ -1 & 1 & 2 \end{bmatrix}$	05	CO2
2(c)	Reduce the quadratic form $3x^2 + 2y^2 + z^2 + 4xy + 4yz$ to the normal form and show that it is positive definite.	05	CO2
3(a)	Find the polar decomposition of $A = \begin{pmatrix} 11 & -5 \\ -2 & 10 \end{pmatrix}$	05	CO3
3(b)	Find the 2-rank approximation of the matrix $A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	05	CO3
3(c)	Find Singular value decomposition of the matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$	05	CO3
4(a)	Verify Green's Lemma in the xy plane for $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the closed curve of the region bounded by $x = 0, y = 0, x + y = 1$.	05	CO4
4(b)	Show that the vector field $\vec{F} = \frac{\vec{r}}{r^3}$ is irrotational as well as solenoidal.	05	CO4

4(c)	If $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$. Evaluate $\iint_S \vec{F} \cdot d\vec{s}$ where S is the surface of the cube $x = 0, x = 1; y = 0, y = 1; z = 0; z = 1$	05	CO4
5(a)	Examine if the set $S = \{(x, y) \in R^2: 3x + 2y \leq 8\}$ is a convex set.	05	CO5
5(b)	If $\frac{3}{x} + \frac{4}{y} + \frac{5}{z} = 6$ find the values of x, y, z which make $x + y + z$ maximum	05	CO5
5(c)	Solve: Min $f(x) = x_1^2 + x_2^2 - 6x_1 - 4x_2 + 13$ Subject to $x_1^2 + x_2^2 \leq 52$ $x_1, x_2 \geq 0$	05	CO5